

The SRDS Formal Framework

A Self-Contained Mathematical Theory of Critical Transitions
in Self-Referential Dissipative Systems

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Abstract

We develop, in full and self-contained form, the mathematical theory of *self-referential dissipative systems* (SRDS): sequences generated by a state recursion whose update operator is itself driven by the state it produces. The class is fixed by five axioms (self-reference, dissipation, accumulation, transition, irreversibility) together with three regularity conditions, and we prove that accumulation, transition, and irreversibility need not be posited as independent axioms: under three explicitly named *constitutive* regularity conditions L1–L3 they follow as *theorems*, with the saturation postulate L2 carrying the transition (the cumulative curve inherits its sigmoid inflection from a damage rate that is a single-peaked function of the accumulated state—the operative form of self-reference), in the manner of a phenomenological constitutive law. On this base we prove three structural results: an information–stability correspondence relating the sign of the maximal Lyapunov exponent to the trend of mutual information (T1); an autopoietic stationarity theorem giving a unique, globally attracting non-equilibrium steady state of the operator recursion (T2); and a Goldilocks principle locating optimal function in a neighbourhood of, rather than exactly at, criticality (T3). We construct the two-scale renormalization operator that maps the dynamics across observational scales, prove the existence and uniqueness of its monotone fixed point through Birkhoff–Hopf projective contraction, and identify the cumulative inflection as the order parameter of the class. Finally we prove a No-Go theorem: within reparametrization-closed operator classes no universal numerical inflection location exists; the symmetric value is recovered exactly when the kernel is mirror-symmetric, and is otherwise a substrate-specific measured quantity. We close with an elimination ledger making precise what cannot be concluded without additional structure, and with the spectral and contraction substructure governing convergence rates. Every result is proved in full, reduced to a cited result of the established literature, or stated under explicitly named additional hypotheses, with the conditions on which it rests made explicit. The framework is metric-agnostic and is instantiated by, but does not depend on, any particular measurement system.

Contents

1	Preliminaries and Notation	1
2	The SRDS Axiomatic Class	3
3	Two-Scale Renormalization	6
4	The Operational Frame	6
5	T1: Information–Stability Correspondence	7

6	T2: Autopoietic Stationarity	9
7	T3: The Goldilocks Principle	10
8	The Renormalization Operator and the Inflection Fixed Point	11
9	The No-Go Theorem	13
10	The Second No-Go: No Static Classifier	15
11	The Elimination Ledger	17
12	Spectral and Contraction Substructure	18

1 Preliminaries and Notation

This section fixes the analytic and information-theoretic vocabulary used throughout and records the standing regularity assumptions. All statements are proved or reduced to standard references.

1.1 Function spaces

Let $C([0, 1])$ denote the space of continuous real functions on $[0, 1]$ with the supremum norm $\|Q\|_\infty := \sup_{x \in [0, 1]} |Q(x)|$.

Proposition 1.1 (Completeness). *($C([0, 1]), \|\cdot\|_\infty$) is a complete metric space.*

Proof. Let $(Q_n)_{n \in \mathbb{N}}$ be Cauchy in $\|\cdot\|_\infty$. For each fixed x , $|Q_n(x) - Q_m(x)| \leq \|Q_n - Q_m\|_\infty$, so $(Q_n(x))_n$ is Cauchy in $\mathbb{R}$ and, by completeness of $\mathbb{R}$, converges to a limit we call $Q(x)$. The convergence is uniform: given $\varepsilon > 0$ choose N with $\|Q_n - Q_m\|_\infty < \varepsilon$ for $n, m \geq N$; letting $m \rightarrow \infty$ in $|Q_n(x) - Q_m(x)| < \varepsilon$ gives $|Q_n(x) - Q(x)| \leq \varepsilon$ for all x and all $n \geq N$, i.e. $\|Q_n - Q\|_\infty \leq \varepsilon$. A uniform limit of continuous functions is continuous: for $n \geq N$ and $|x - x_0|$ small enough that $|Q_n(x) - Q_n(x_0)| < \varepsilon$ (continuity of Q_n), the triangle inequality $|Q(x) - Q(x_0)| \leq |Q(x) - Q_n(x)| + |Q_n(x) - Q_n(x_0)| + |Q_n(x_0) - Q(x_0)| < 3\varepsilon$ holds. Hence $Q \in C([0, 1])$ and $Q_n \rightarrow Q$. $\square$

Let $BV([0, 1])$ denote the functions of bounded variation, with total variation

$$\text{Var}(Q) := \sup_P \sum_i |Q(x_i) - Q(x_{i-1})|$$

over partitions $P = \{0 = x_0 < \dots < x_k = 1\}$.

Proposition 1.2. *Every $Q \in BV([0, 1])$ is bounded, with $|Q(x)| \leq |Q(0)| + \text{Var}(Q)$ for all x .*

Proof. For fixed x the partition $\{0, x\}$ gives $|Q(x) - Q(0)| \leq \text{Var}(Q)$, whence $|Q(x)| \leq |Q(0)| + \text{Var}(Q)$. $\square$

We write $\mathcal{M} \subset BV([0, 1])$ for the set of *anchored monotone profiles*: nonincreasing Q with $Q(0) = 1$ (the *survival* convention, required by the projective gauge of Section 8: the anchored endpoint must be strictly positive, which $Q(0) = 1$ supplies and a CDF anchor $Q(0) = 0$ would not; the second anchor $Q(1) = 0$ is imposed by postprocessing where a two-end normalization is needed). Such a Q is the survival function $1 - F$ of a probability measure F on $[0, 1]$; the cumulative-damage curve $s(t)$ of Section 2 is the complementary CDF $1 - Q$, with the *same* inflection location $\hat{a}$. By Proposition 1.2, $\mathcal{M}$ is uniformly bounded.

1.2 Information theory

Definition 1.3. For a discrete random variable X with law p , the Shannon entropy is $H(X) := -\sum_x p(x) \log p(x)$ (convention $0 \log 0 = 0$). For a pair (X, Y) the joint entropy is $H(X, Y) := -\sum_{x,y} p(x, y) \log p(x, y)$, the conditional entropy $H(X | Y) := H(X, Y) - H(Y)$, and the mutual information $I(X; Y) := H(X) + H(Y) - H(X, Y)$.

Lemma 1.4 (Properties of mutual information). *I satisfies: (i) $I(X; Y) \geq 0$; (ii) $I(X; Y) = I(Y; X)$; (iii) $I(X; Y) = 0$ iff X, Y are independent; (iv) $I(X; Y) \leq \min\{H(X), H(Y)\}$.*

Proof. Write $I(X; Y) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$. (i) By Jensen's inequality applied to the convex function $t \mapsto t \log t$, equivalently by the nonnegativity of the Kullback–Leibler divergence $D(p(x, y) \| p(x)p(y)) = I(X; Y) \geq 0$ [8]. (ii) Symmetry is immediate from $I = H(X) + H(Y) - H(X, Y)$. (iii) Equality $I = 0$ holds iff $D = 0$, i.e. $p(x, y) = p(x)p(y)$ everywhere, which is independence. (iv) $I(X; Y) = H(X) - H(X | Y)$ and $H(X | Y) \geq 0$ for discrete variables give $I \leq H(X)$; symmetry gives $I \leq H(Y)$. $\square$

Theorem 1.5 (Data-processing inequality). *If $X \rightarrow Y \rightarrow Z$ is a Markov chain then $I(X; Z) \leq I(X; Y)$.*

Proof. Standard; see [8, Thm. 2.8.1]. Processing of Y cannot increase information about X . $\square$

For continuous laws we use the differential entropy $h(X) := -\int p \log p$ and the Gaussian maximum-entropy bound: among densities on $\mathbb{R}^n$ with covariance Σ , the Gaussian maximizes h , with $h(X) \leq \frac{1}{2} \log((2\pi e)^n \det \Sigma)$ [8, Thm. 8.6.5].

1.3 The measurement model

The framework is metric-agnostic: it constrains the *geometry of a trajectory*, not the instrument that records it. We fix the minimal measurement vocabulary.

Definition 1.6 (Specimen and deformation curve). A *specimen* is a finite ordered sequence $(z_1, \dots, z_T)$ of states $z_t \in \mathbb{R}^m$, obtained from an embedding map φ of the raw sequence into $\mathbb{R}^m$. The index t is ordered loading time, not an i.i.d. sample index; the ordered sequence is the *deformation curve* of the specimen. For window size W and $t \geq W$ the window is $\mathcal{W}(t) := \{t - W + 1, \dots, t\}$ (with $\mathcal{W}(t) := \{1, \dots, t\}$ for $t < W$).

Because the index is loading time, successive measurements along a deformation curve are *autocorrelated by construction*; this is signal, not nuisance, and inference at the specimen level must treat the whole curve as the unit of analysis (Section 4).

1.4 Standing regularity assumptions

The dynamical results invoke the following conditions. They are sufficient, not necessary, and each is satisfied by standard generative models of ordered embedded sequences.

Assumption R1 (Stochastic flow). The state sequence (z_t) is generated by a discrete-time process whose continuous-time interpolation solves $dz = F(z) dt + \sigma dW_t$, with $F : \mathbb{R}^m \rightarrow \mathbb{R}^m$ locally Lipschitz, $\sigma > 0$, and W_t standard Brownian motion in $\mathbb{R}^m$.

Assumption R2 (Lyapunov spectrum). Along trajectories the variational equation $\dot{\xi} = DF(z(t)) \xi$ admits a maximal Lyapunov exponent $\lambda_{\max} := \lim_{t \rightarrow \infty} \frac{1}{t} \log \|\xi(t)\| / \|\xi(0)\|$, which exists for a.e. initial condition by the multiplicative ergodic theorem [19].

Assumption R3 (Bi-Lipschitz embedding). On the support of the trajectory law, φ is bi-Lipschitz: there are $0 < c_1 \leq c_2 < \infty$ with $c_1 d_{\mathcal{X}}(s, s') \leq \|\varphi(s) - \varphi(s')\| \leq c_2 d_{\mathcal{X}}(s, s')$ for a base metric $d_{\mathcal{X}}$.

Assumption R4 (Ergodicity). (z_t) is stationary and ergodic with invariant law μ having log-concave density on $\mathbb{R}^m$; hence time averages converge to μ -averages [3] and concentration holds [5].

Assumption R5 (Finite information). For every t and every finite measurable partition, the discretized state coordinates have finite Shannon entropy, so $I(\cdot; \cdot)$ is well defined.

2 The SRDS Axiomatic Class

2.1 Two recursions

Definition 2.1 (Self-referential system). A self-referential dissipative system is a pair of coupled recursions

$$z_{t+1} = F_t(z_t, u_t), \quad F_{t+1} = \Phi(F_t, z_t, u_t), \quad (1)$$

where $z_t \in \mathbb{R}^m$ is the state, u_t an exogenous input, $F_t \in \mathcal{F}$ the update operator drawn from a complete metric space $(\mathcal{F}, d_{\mathcal{F}})$, and $\Phi : \mathcal{F} \times \mathbb{R}^m \times \mathcal{U} \rightarrow \mathcal{F}$ the meta-operator. A configuration F^* is *autopoietic* if $\Phi(F^*, z^*, u^*) = F^*$.

The first recursion is classical feedback dynamics [28, 1]. It is insufficient alone: a specimen remembers its loading history (the Bauschinger effect), a sequence carries path dependence. The second recursion makes that memory explicit: the operator that writes the next state is itself rewritten during writing. This is the operative closure of [17].

2.2 The five axioms

We collect the cumulative dissipation along a trajectory. Under Assumption R1 let $V : \mathcal{F} \rightarrow [0, \infty)$ be a measurable functional and define the one-step expected decrement and the cumulative damage

$$\delta_t := V(F_t) - \mathbb{E}[V(F_{t+1}) \mid \mathcal{G}_t], \quad \Sigma_t := \sum_{s < t} \delta_s, \quad (2)$$

where $(\mathcal{G}_t)$ is the natural filtration.

Axiom A0 (Self-reference). The update operator evolves by the second recursion of (1): $F_{t+1} = \Phi(F_t, z_t, u_t)$; the operator is itself a state, rewritten as it acts.

Axiom A1 (Dissipation). There is a Lyapunov functional $V \geq 0$ with $\mathbb{E}[V(F_{t+1}) \mid \mathcal{G}_t] \leq V(F_t)$, i.e. $\delta_t \geq 0$ in conditional expectation. Equivalently the system exports entropy to maintain bounded internal organization far from equilibrium [21].

Axiom A2 (Accumulation). The damage Σ_t is monotone nondecreasing: $\Sigma_{t+1} \geq \Sigma_t$.

Axiom A3 (Transition). The normalized damage $\sigma(t) := \Sigma_t / \Sigma_{\infty}$ admits a unique inflection point $\hat{a}$ (sigmoid form), the order parameter of the trajectory [24].

Axiom A4 (Irreversibility). A trajectory that has crossed the inflection ($\Sigma > \hat{a}$) returns to its pre-transition configuration with probability strictly less than one [25].

2.3 Regularity conditions

Regularity Condition L1 (Lyapunov half-order). V is bounded below (normalized $V \geq 0$) and adapted to $(\mathcal{G}_t)$, so the increments δ_t are nonnegative adapted random variables and Σ_t is a nondecreasing adapted process.

Regularity Condition L2 (Constitutive damage-rate law). The mean cumulative damage $s(t) := \mathbb{E}[\Sigma_t]/\mathbb{E}[\Sigma_\infty] \in [0, 1]$ evolves by an *autonomous* law in which the dissipation rate is a function of the accumulated state, $\dot{s} = g(s)$, with $g : [0, 1] \rightarrow [0, \infty)$ continuous and: (a) $g(s) > 0$ on $(0, 1)$ (damage accrues while unsaturated); (b) $g(1) = 0$ (saturation: the rate vanishes at the autopoietic configuration); (c) g is single-peaked — there is $s^* \in (0, 1)$ with $g' > 0$ on $(0, s^*)$ and $g' < 0$ on $(s^*, 1)$ (early self-amplification, late saturation). The canonical instance is the logistic $g(s) = r s(1 - s)$. That g depends on the accumulated state s is the operative content of self-reference (Axiom A0) on the damage coordinate; (b) is dissipation (Axiom A1). Finiteness $\mathbb{E}[\Sigma_\infty] \leq V(F_0)$ follows from $V \geq 0$ in L1.

Regularity Condition L3 (Asymmetry measure). The stochastic-thermodynamic entropy production $\Delta\Sigma$ along any segment satisfies a detailed fluctuation relation $\mathbb{P}[\gamma^\dagger]/\mathbb{P}[\gamma] = e^{-\Delta\Sigma}$, where $\gamma^\dagger$ is the time-reversed segment of γ [9, 25].

2.4 The reduction theorem

The point of the axiomatization is that only A0–A1 are primitive: accumulation, transition, and irreversibility are *theorems* once the regularity conditions hold. Making the reduction explicit lets the load-bearing capacity of each axiom be tested on each substrate separately, and it is exactly this structure that produces the predicted null findings of a substrate that violates one regularity condition. These conditions are *constitutive postulates*, not merely technical hypotheses: in particular L2 is the load-bearing assumption from which the transition follows. The reduction is accordingly a *consistency* result—given the named postulates, accumulation, transition and irreversibility are not independent axioms—rather than a derivation from self-reference and dissipation alone.

Theorem 2.2 (Reduction). *Under Axioms A0–A1:*

- (i) (Accumulation) *L1+L2 imply A2, and $\Sigma_t \uparrow \Sigma_\infty < \infty$.*
- (ii) (Transition, in expectation) *L2 implies the mean form of A3: the expected cumulative profile $s(t) = \mathbb{E}[\Sigma_t]/\mathbb{E}[\Sigma_\infty]$ is sigmoid, with a unique inflection at the loading time $\hat{a}$ at which the damage rate peaks, $s(\hat{a}) = s^*$ — provided the onset satisfies $s(0) < s^*$; if $s(0) \geq s^*$ the profile is purely concave (inflection at the origin: the “measured beyond yield” boundary case).*
- (iii) (Irreversibility) *L3 implies A4.*

Proof. (i) By Axiom A1 the conditional increments satisfy $\delta_t \geq 0$; by L1 they are nonnegative and adapted, so $\Sigma_{t+1} = \Sigma_t + \delta_t \geq \Sigma_t$ pathwise, which is Axiom A2; thus Σ_t is the predictable compensator of the nonnegative supermartingale $V(F_t)$. Taking expectations and telescoping, $\mathbb{E}[\Sigma_t] = \sum_{s < t} \mathbb{E}[\delta_s] = V(F_0) - \mathbb{E}[V(F_t)] \leq V(F_0) < \infty$, using $V \geq 0$ (L1). A nondecreasing process with uniformly bounded expectation converges almost surely by the monotone convergence theorem to a limit Σ_∞ with $\mathbb{E}[\Sigma_\infty] \leq V(F_0) < \infty$; hence $\Sigma_\infty < \infty$ a.s.

(ii) By L2 the mean damage $s(t) = \mathbb{E}[\Sigma_t]/\mathbb{E}[\Sigma_\infty]$ — the deterministic profile actually fitted — solves the autonomous equation $\dot{s} = g(s)$ from an onset $s(0) = s_0 \in (0, 1)$. Since $g > 0$ on $(0, 1)$, s is strictly increasing, and $g(1) = 0$ gives $s(t) \rightarrow 1$; the profile rises from s_0 to 1. Its curvature is

$\ddot{s} = \frac{d}{dt}g(s) = g'(s)g(s)$, so $\text{sign}(\ddot{s}) = \text{sign}(g'(s))$ because $g > 0$ on the interior. By the single-peak property L2(c), g' changes sign exactly once, from $+$ to $-$, at s^* . If $s_0 < s^*$, strict monotonicity makes s attain s^* at exactly one time $\hat{a}$: s is convex on $[0, \hat{a})$ and concave on $(\hat{a}, 1)$, i.e. sigmoid with a unique inflection at $\hat{a}$, the loading time of maximal expected damage rate — the yield point (Axiom A3). If instead $s_0 \geq s^*$, the trajectory meets only $g' < 0$, so $\ddot{s} < 0$ throughout: the profile is purely concave with no interior inflection (a substrate measured beyond its yield onset). Crucially the inflection is *derived* from the state-dependence of g — read off the accumulated level s^* at which the rate peaks — rather than *posited* as a unimodality in time; this is what removes the circularity of inferring the inflection from an assumed time profile. The statement is at the level of the expected trajectory; the pathwise inflection of an individual realization, $\sigma(t) = \Sigma_t/\Sigma_\infty$ as in Axiom A3, is the empirically fitted quantity (one $\hat{a}$ per specimen) and agrees with the mean inflection under regularity of the increment process, but is not asserted here as a pathwise theorem.

(iii) Call a trajectory a *crossing* if $\Sigma > \hat{a}$ at some time, and write $\mathcal{C}$ for the set of crossings. The return event $\mathcal{R}$ —that the time-reversed dynamics restores the pre-transition configuration—is realized precisely by the reversals $\{\gamma^\dagger : \gamma \in \mathcal{C}\}$: a return is, by definition, a forward crossing carried back. On a crossing the damage rate is strictly positive ($g > 0$ on $(0, 1)$ by L2), so the accumulated entropy production satisfies $\Delta\Sigma(\gamma) > 0$ strictly for every $\gamma \in \mathcal{C}$. By L3, $\mathbb{P}[\gamma^\dagger] = e^{-\Delta\Sigma(\gamma)} \mathbb{P}[\gamma]$, hence

$$\mathbb{P}[\mathcal{R}] = \sum_{\gamma \in \mathcal{C}} \mathbb{P}[\gamma^\dagger] = \sum_{\gamma \in \mathcal{C}} e^{-\Delta\Sigma(\gamma)} \mathbb{P}[\gamma] \leq \sum_{\text{all } \gamma} e^{-\Delta\Sigma(\gamma)} \mathbb{P}[\gamma] = \mathbb{E}[e^{-\Delta\Sigma}] = 1,$$

the inequality holding because $\mathcal{C}$ is a subset of all paths and every summand $e^{-\Delta\Sigma} \mathbb{P}[\gamma] \geq 0$, and the last equality being the integral fluctuation theorem (the detailed relation L3 summed over all paths) [25]. The inequality is *strict*: under Assumption R1 the non-crossing paths (the diffusion staying below $\hat{a}$ over a finite window) carry positive probability and each contributes $e^{-\Delta\Sigma} \mathbb{P}[\gamma] > 0$ to the full sum but nothing to $\mathbb{P}[\mathcal{R}]$; hence $\mathbb{P}[\mathcal{R}] < 1$, which is Axiom A4. (The naive bound “ $\Delta\Sigma > 0$ with positive probability $\Rightarrow \mathbb{E}[e^{-\Delta\Sigma}] < 1$ ” is invalid, since reversed paths can carry $\Delta\Sigma < 0$; the integral fluctuation theorem is what makes the argument rigorous.) Self-reference (A0) is what makes $\gamma^\dagger \neq \gamma^{-1}$: because F_t carries the history, the reverse state path is not generated by inverting the forward operator, so the entropy production is genuine rather than a reparametrization artifact. $\square$

Remark 2.3 (Nulls as violated regularity). Theorem 2.2 predicts its own failures. A substrate whose recovery balances damage ($\mathbb{E}[\delta_t] \rightarrow 0$, violating the strict-rise part of L2) shows no transition; a symmetric self-reference ($\gamma^\dagger = \gamma^{-1}$, violating L3) shows no irreversibility; an externally imposed sequence with no path asymmetry violates L3 as well. In each case the absence of a critical transition is a confirmed prediction of the reduction, not a counterexample to it.

3 Two-Scale Renormalization

A self-referential system is observed at several scales — token, turn, sequence, corpus, population — and the framework requires that the same structural law describe each. We make the inter-scale map precise.

Definition 3.1 (Renormalization map). For scales $\ell < \ell'$ the renormalization map is the composition

$$M_{\ell \rightarrow \ell'} = N_{\ell'} \circ R_{\ell \rightarrow \ell'} \circ C_{\ell \rightarrow \ell'}, \quad (3)$$

where C *coarse-grains* micro-trajectories into blocks (windows, clusters, representative volumes), R *rescales* the effective parameters along the induced parameter flow $\theta_\ell \mapsto \theta_{\ell'}$, and N *normalizes*

dimensionless invariants so that the renormalized description stands structurally on a par with the original.

Definition 3.2 (Normal form and scale-covariance ansatz). We take the mean dynamics in the Onsager-type normal form

$$\dot{z} = F(z, \Sigma) - \nabla_z \Pi(z, \Sigma) + \Theta \nabla_z I(z), \quad (4)$$

with drift F , dissipation potential Π , and information coupling I : the gradient form of Assumption R1 in which $-\nabla_z \Pi$ is the dissipation of Axiom A1 and $\Theta \nabla_z I$ is the self-referential force of Axiom A0. We restrict attention to the *relevant sector*: the subspace of operators on which $M_{\ell \rightarrow \ell'}$ acts form-covariantly, returning (4) with renormalized $(F', \Pi', \Theta') = R_{\ell \rightarrow \ell'}(F, \Pi, \Theta)$.

Remark 3.3 (On the ansatz). Renormalization generically generates new operators, so exact form-closure is not automatic; the restriction to the relevant sector is the standard content of a renormalization-group analysis, in which irrelevant operators are suppressed under coarse-graining and the surviving form-covariant subspace carries the universal structure [14, 29]. Within this sector the construction is consistent: coarse-graining C replaces z by a block average whose fluctuations contribute, to leading order, a renormalized diffusion absorbed into Π' (Assumption R4), and each gradient term pulls back to a gradient term under the smooth change of variables induced by R . Downstream results use only this covariance, not a claim that every microscopic dynamics lies in the relevant sector.

The five operative scales are: tokens (ℓ_0), turn-pairs (ℓ_1), the sequence (ℓ_2 , carrier of the inflection $\hat{a}$ and the accumulator), the corpus (ℓ_3 , carrier of the $\hat{a}$ -distribution), and the population (ℓ_4). The scale-covariance of Definition 3.2 is what licenses transporting the order parameter $\hat{a}$ across them.

4 The Operational Frame

The abstract class of Section 2 is instantiated, at the sequence scale ℓ_2 , by a measurement system on embedded trajectories. The instantiation is metric-agnostic in principle; we record the current operational definitions so that the abstract objects $(\Sigma, \hat{a})$ have explicit observable counterparts.

Definition 4.1 (Deformation observables). Let $(z_t)_{t \leq T}$, $z_t = (u_t, a_t) \in \mathbb{R}^{2d}$, be the embedded turn-pair sequence and $v_t := \|z_t - z_{t-1}\|$ the semantic velocity. Five coupled deformation axes are defined on windows $\mathcal{W}(t)$:

- (D1) **Stiffness** $E_a := 1/\text{Var}_{\mathcal{W}}(v)$, inverse velocity variance (strain hardening);
- (D2) **Asymmetry** MAS, the meta-asymmetry between the user and assistant sub-trajectories in $\mathbb{R}^d$ (one-sided load introduction, cantilever);
- (D3) **Accumulation** B_w , the windowed budget accumulator, a monotone functional of the cumulative geometric displacement (creep/fatigue);
- (D4) **Brittleness** dir_consistency , the directional coherence of successive increments, whose collapse marks brittle break-out;
- (D5) **Coupling** $\text{xcorr}(v, w)$, the cross-correlation of semantic velocity v and a structural displacement field w (dynamics–structure covariation).

The order parameter is the **inflection** $\hat{a}$: the location of the unique inflection (Theorem 2.2(ii)) of the sigmoid fit of cumulative damage Σ against cumulative strain.

Remark 4.2 (Anti-sparse coupling). The five axes are coupled, not redundant. In the linear-response (Hooke) regime $\sigma = E\varepsilon$ ties stress, stiffness and strain analytically; the same coupling recurs among the axes, so stiffness (D1) and accumulation (D3) are two projections of one material response. A sparsity-inducing (ℓ_1) selection would discard such coupled coordinates as collinear and thereby lose information, not noise: the correct paradigm is materials testing, where coupled gauges record one response from distinct axes, not feature selection over independent predictors. Consistently, the loading-time autocorrelation of the cumulative family is signal: measurements along a deformation curve are not independent, and specimen-level inference must use the whole curve as the unit (block resampling), not the individual measurement.

Definition 4.3 (Budget state machine). The accumulator B_w drives a monotone hysteretic state machine with ordered regimes $\text{NORMAL} \preceq \text{WATCH} \preceq \text{WARNING} \preceq \text{CRITICAL}$, transitions of which are governed by policy thresholds and a recovery rate. By the No-Go theorem (Section 9) these thresholds are calibration parameters, not universal constants; only the *architecture* (monotone accumulation with hysteretic, asymmetric transitions, Axioms A2,A4) is part of the theory.

The embedding instrument is a sentence encoder satisfying Assumption R3 [22]; decreasing embedding variance signals stiffening of the semantic space, and irreversibly growing cumulative displacement documents plastic deformation, the observable face of Axioms A2 and A4.

5 T1: Information–Stability Correspondence

The first structural theorem links dynamical stability, measured by the maximal Lyapunov exponent, to the trend of mutual information between the coupled coordinates $p := u$ (prompt/user) and $r := a$ (response/assistant), $z = (p, r)$.

Lemma 5.1 (Covariance growth). *Under Assumptions R1–R3, in the transient regime—before the process settles to the invariant law of Assumption R4—if $\lambda_{\max} > 0$ then the state covariance grows exponentially along the leading direction, $\|\text{Cov}(t)\|_{\text{op}} \geq c e^{2\lambda_{\max} t}$ for some $c > 0$ over the transient window. (At stationarity Cov is time-invariant; the exponential growth is a transient, not an asymptotic, statement and does not contradict R4.)*

Proof. By Assumption R1 a perturbation obeys the variational equation $\dot{\xi} = DF(z(t))\xi$. The covariance solves the Lyapunov equation $\dot{\text{Cov}} = DF \text{Cov} + \text{Cov} DF^\top + \sigma^2 I$. For the homogeneous part the solution is $\text{Cov}_h(t) = \Phi(t) \text{Cov}(0) \Phi(t)^\top$ with fundamental matrix Φ . By Assumption R2 and the multiplicative ergodic theorem [19], $\frac{1}{t} \log \|\Phi(t)\|_{\text{op}} \rightarrow \lambda_{\max}$, so choosing $\text{Cov}(0)$ with a component along the leading Oseledets direction gives $\|\text{Cov}_h(t)\|_{\text{op}} \geq c e^{2\lambda_{\max} t}$. The inhomogeneous noise term contributes $\int_0^t \Phi(t) \Phi(s)^{-1} \sigma^2 \Phi(s)^{-\top} \Phi(t)^\top ds$, which is positive semidefinite and of order at most $e^{2\lambda_{\max} t}$; it cannot reduce the leading growth. Hence the stated lower bound. $\square$

Lemma 5.2 (Two-sided entropy bounds). *Let $X \in \mathbb{R}^n$ have covariance Σ_X . Then $h(X) \leq \frac{1}{2} \log((2\pi e)^n \det \Sigma_X)$, with equality iff X is Gaussian [8, Thm. 8.6.5]. If moreover the law of X is log-concave (Assumption R4), there is a dimensional constant $c_n > 0$ with $h(X) \geq \frac{1}{2} \log \det \Sigma_X + c_n$.*

Proof. The upper bound is the Gaussian maximum-entropy property. For the lower bound, reduce to the isotropic case by $Y := \Sigma_X^{-1/2}(X - \mathbb{E}X)$, which is log-concave with identity covariance and satisfies $h(X) = h(Y) + \frac{1}{2} \log \det \Sigma_X$. Isotropic log-concave laws obey a uniform entropy lower

bound $h(Y) \geq c_n$ by the variance–entropy comparison from the Brascamp–Lieb inequality and the concentration of log-concave measures [6, 5]. Combining gives the claim. $\square$

Theorem 5.3 (T1: Information–stability correspondence). *Assume R1–R5 together with the coupling non-degeneracy condition (N): in the expanding regime the cross-covariance grows strictly slower than the geometric mean of the marginal blocks, $\|\Sigma_{pr}(t)\| = o(\sqrt{\|\Sigma_{pp}(t)\| \|\Sigma_{rr}(t)\|})$, and in the contracting regime the conditional law of r given p has a bounded noise floor. Then, in the transient regime (the approach to the invariant law of R4), with $I(t) := I(p_t; r_t)$,*

$$\text{sign}(\lambda_{\max}) = -\text{sign}\left(\frac{d}{dt}I(t)\right) :$$

instability ($\lambda_{\max} > 0$) forces coupling decay and stability ($\lambda_{\max} < 0$) forces coupling growth; once the process reaches the stationary law, Cov and hence I are time-invariant ($\lambda_{\max} = 0$, $dI/dt = 0$). The correspondence is therefore a transient diagnostic, not an identity asserted at the stationary law.

Proof. Write $I = h(p) + h(r) - h(p, r)$.

Case $\lambda_{\max} > 0$. By Lemma 5.1 the state covariance expands at rate $e^{2\lambda_{\max}t}$ along the leading Oseledets direction. The conclusion does not require every block to share this rate: condition (N) directly forces the off-diagonal block to grow strictly slower than the geometric mean of the diagonal blocks, so the correlation coefficient $\rho_{pr} = \Sigma_{pr}/\sqrt{\Sigma_{pp}\Sigma_{rr}} \rightarrow 0$ regardless of which marginal carries the expansion. As $\rho_{pr} \rightarrow 0$ the joint law tends, in the log-concave leading order of Lemma 5.2, to the product of the marginals, so $I(p_t; r_t) \downarrow 0$; the mutual-information trend is therefore strictly negative.

Case $\lambda_{\max} < 0$. By Assumption R4 the covariance is bounded (contraction toward the invariant law). Contraction merges nearby trajectories, so the dependence $r = g(p) + \xi$ sharpens toward the bounded noise floor of (N): $h(r | p)$ decreases toward the noise-floor entropy while $h(r)$ stays bounded below. Then $I = h(r) - h(r | p)$ is nondecreasing, and strictly increasing while the system still approaches stationarity, i.e. $dI/dt \geq 0$, with $dI/dt > 0$ off stationarity.

Case $\lambda_{\max} = 0$. Neither growth nor contraction dominates; the covariance and the conditional structure are asymptotically constant, so I reaches a stationary value and $dI/dt = 0$. This is exactly the boundary at which the sign of dI/dt changes.

Collecting the three cases, $\text{sign}(\lambda_{\max}) = -\text{sign}(dI/dt)$. $\square$

Remark 5.4 (Excluded degeneracy). Condition (N) is necessary, not cosmetic. If a single unstable mode drives both blocks at once (the leading Oseledets direction projecting onto p and r with comparable weight), then $\rho_{pr} \rightarrow 1$ and instability *increases* the coupling, reversing the correspondence. (N) excludes exactly this synchronization; it holds whenever the two channels are not slaved to one common expanding mode.

Remark 5.5. T1 is a diagnostic: a declining mutual-information trend signals dynamical instability without computing Lyapunov exponents, and growing or stable coupling signals stability or criticality. It is the abstract content of axis (D5) of Definition 4.1.

6 T2: Autopoietic Stationarity

The second theorem gives conditions under which the operator recursion of Definition 2.1 has a unique, globally attracting non-equilibrium steady state. It requires one structural hypothesis beyond the axioms.

Definition 6.1 (Contraction condition). The meta-operator Φ acts on a complete metric space $(\mathcal{F}, d_{\mathcal{F}})$ as a κ -contraction in its first argument: there is $\kappa \in (0, 1)$ with $d_{\mathcal{F}}(\Phi(F_1, \cdot), \Phi(F_2, \cdot)) \leq \kappa d_{\mathcal{F}}(F_1, F_2)$ for all F_1, F_2 . Moreover the stationary state drift is *two-sided dissipative* about its mean z^* : there exist $0 < c \leq c' < \infty$ with $-c' \|z - z^*\|^2 \leq \langle z - z^*, F^*(z) \rangle \leq -c \|z - z^*\|^2$ on the basin. (This last is a hypothesis on the *state* drift; Axiom A1 concerns the operator-level functional and does not by itself bound F^* .)

Theorem 6.2 (T2: Autopoietic stationarity). *Under Axioms A0–A1 and Definition 6.1:*

- (i) *There is a unique autopoietic fixed point F^* with $\Phi(F^*) = F^*$, and the iterates $F_{n+1} = \Phi(F_n)$ converge geometrically, $d_{\mathcal{F}}(F_n, F^*) \leq \frac{\kappa^n}{1-\kappa} d_{\mathcal{F}}(F_1, F_0)$.*
- (ii) *The state dynamics under F^* possess a globally attracting non-equilibrium regime whose stationary fluctuation level is bounded away from zero, in the asymptotically absorbing band $[\sigma^2 m / (4c'), \sigma^2 m / (4c)]$.*
- (iii) *At F^* the dissipation functional is stationary, $\dot{D} = 0$.*

Proof. (i) $(\mathcal{F}, d_{\mathcal{F}})$ is complete and Φ is a κ -contraction (Definition 6.1); the Banach fixed-point theorem [2] yields a unique F^* with $\Phi(F^*) = F^*$ and, for any F_0 , the a-priori estimate $d_{\mathcal{F}}(F_n, F^*) \leq \frac{\kappa^n}{1-\kappa} d_{\mathcal{F}}(F_1, F_0)$.

(ii) Fix the operator at F^* ; the state then solves the autonomous diffusion $dz = F^*(z) dt + \sigma dW_t$ (Assumption R1) with stationary mean $z^* = \mathbb{E}_{\mu}[z]$. Define the quadratic Lyapunov functional $\mathcal{V}(z) := \frac{1}{2} \|z - z^*\|^2$. By Itô's formula,

$$d\mathcal{V} = \langle z - z^*, F^*(z) \rangle dt + \frac{1}{2} \sigma^2 m dt + \sigma \langle z - z^*, dW_t \rangle.$$

The two-sided dissipativity of Definition 6.1 gives $-2c' \mathcal{V}(z) \leq \langle z - z^*, F^*(z) \rangle \leq -2c \mathcal{V}(z)$. Write $\bar{V}(t) := \mathbb{E}[\mathcal{V}(z(t))]$; taking expectations and using the Itô noise injection rate $\frac{1}{2} \sigma^2 m$, where m is the state dimension of Assumption R1, the two bounds sandwich the mean Lyapunov rate:

$$-2c' \bar{V} + \frac{1}{2} \sigma^2 m \leq \dot{\bar{V}} \leq -2c \bar{V} + \frac{1}{2} \sigma^2 m.$$

Both scalar comparison ODEs are globally attracting (to $\sigma^2 m / (4c')$ and $\sigma^2 m / (4c)$ respectively), so by comparison

$$\liminf_{t \rightarrow \infty} \bar{V}(t) \geq \frac{\sigma^2 m}{4c'} > 0, \quad \limsup_{t \rightarrow \infty} \bar{V}(t) \leq \frac{\sigma^2 m}{4c}.$$

The band $[\sigma^2 m / (4c'), \sigma^2 m / (4c)]$ is asymptotically absorbing (the limit exists when $c = c'$). The lim inf bound keeps the stationary fluctuation level bounded away from 0 (the system maintains finite fluctuations rather than collapsing to a deterministic point, non-equilibrium); absorption into the band is global attraction, stability.

(iii) Let $D := \mathbb{E}[C(z, u)] - C_{\text{eq}}$ be the dissipation functional, with C the cost whose minimization is the gradient structure of (4). Then $\dot{D} = \partial_t \mathbb{E}[C]$. In the stationary regime of (ii) the law of z is time-invariant, so $\partial_t \mathbb{E}[C] = 0$; equivalently z^* is a critical point of C along F^* , $\langle \nabla_z C(z^*), F^*(z^*) \rangle = 0$. Hence $\dot{D} = 0$: entropy production is exactly balanced by dissipative export, the defining balance of a non-equilibrium steady state [21]. $\square$

Remark 6.3. T2 is the existence statement behind the autopoietic configuration of Definition 2.1: self-reference plus dissipation plus contraction produces a stable far-from-equilibrium attractor, not a dead equilibrium. The geometric rate in (i) is conditional on a *uniform* modulus $\kappa < 1$; if the per-step contraction degrades ($\kappa_n \rightarrow 1$), convergence is governed by the cumulative budget rather than a fixed rate (Section 12).

7 T3: The Goldilocks Principle

The third result concerns where, relative to criticality, a self-referential system functions best. It is stated as a hypothesis, with a proved partial result; the hypothesis status is intrinsic, because the performance functional is not fixed by the axioms.

Hypothesis 7.1 (Goldilocks principle). Let $L(I, C)$ be a performance functional in innovation I (exploration/variety) and coherence C (stability/recurrence), jointly concave with positive cross-derivative $\partial^2 L / \partial I \partial C > 0$. Then optimal performance requires interior balance, neither $I = 0$ nor $C = 0$, and the optimum lies in a neighbourhood of criticality $|\lambda_{\max}| \approx 0$ rather than at the measure-zero point of exact criticality.

Proposition 7.2 (Necessity of interior balance). Let $L : [0, \infty)^2 \rightarrow \mathbb{R}$ be C^2 and strictly concave, with positive cross-partial $\partial^2 L / \partial I \partial C > 0$ and strictly positive boundary gradients $\partial_I L(0, 0) > 0$, $\partial_C L(0, 0) > 0$. If L attains a maximum on $[0, \infty)^2$, it is at a unique interior point (I^*, C^*) with $I^* > 0$, $C^* > 0$, and every boundary point is strictly suboptimal.

Proof. A strictly concave function on a convex domain has at most one local maximum, which is then global. At an interior maximum the first-order conditions $\partial_I L = \partial_C L = 0$ hold. Suppose the maximum were on the boundary, say at $I = 0$; optimality on that face requires $\partial_I L \leq 0$ along $I = 0$. But by the positive cross-partial and the strict boundary-gradient hypotheses, for every $C \geq 0$,

$$\partial_I L(0, C) = \partial_I L(0, 0) + \int_0^C \partial_{IC}^2 L dC' \geq \partial_I L(0, 0) > 0,$$

since the integrand is positive (the integral is ≥ 0 , vanishing only at the corner $C = 0$). This contradicts boundary optimality on the entire face $I = 0$, the corner $(0, 0)$ included; the maximum is therefore interior with $I^* > 0$, and symmetrically $C^* > 0$. Strict concavity makes the interior critical point the unique global maximum, with every boundary point strictly below it. $\square$

Remark 7.3 (Why a hypothesis). Proposition 7.2 proves the interior-balance *structure*; the full Hypothesis 7.1 remains open because (a) the functional L is application-dependent and not derivable from the axioms, and (b) the identification of the interior optimum with the neighbourhood of $\lambda_{\max} = 0$ is an empirical prediction: systems operating near, but not at, criticality outperform both rigid (C saturated, $\lambda_{\max} \ll 0$) and chaotic ($\lambda_{\max} \gg 0$) regimes. The principle is the optimization counterpart of T1: criticality is the boundary where the information–stability sign flips, and function concentrates in its neighbourhood.

8 The Renormalization Operator and the Inflection Fixed Point

We now realize the inflection $\hat{a}$ of Axiom A3 as the order parameter of a concrete operator fixed point, and quantify its convergence. We make the construction self-contained.

8.1 The cone, the smoothing channel, and strict positivity

Let

$$\mathcal{K} := \{f \in C([0, 1]) : f \geq 0, f \text{ non-increasing}\}, \quad \mathcal{M} := \{Q \in \mathcal{K} : Q(0) = 1\}$$

be the closed convex cone of non-negative non-increasing profiles and its anchored subset (no constraint $Q(1) = 0$ is imposed). For strictly positive $f, g \gg 0$ the Hilbert projective metric is

$$d_H(f, g) := \ln \sup_{x \in [0, 1]} \frac{f(x)}{g(x)} - \ln \inf_{x \in [0, 1]} \frac{f(x)}{g(x)},$$

which is scale-invariant: $d_H(\alpha f, \beta g) = d_H(f, g)$ for $\alpha, \beta > 0$. A kernel $k : [0, 1]^2 \rightarrow (0, \infty)$ is *totally positive of order two* (TP2) if $k(x_1, y_1)k(x_2, y_2) \geq k(x_1, y_2)k(x_2, y_1)$ for $x_1 < x_2, y_1 < y_2$; the normalized smoothing operator is $(T_\varepsilon f)(x) := \int_0^1 \tilde{k}_\varepsilon(x, y) f(y) dy$ with $\tilde{k}_\varepsilon(x, y) = k_\varepsilon(x, y)/Z_\varepsilon(x)$, $Z_\varepsilon(x) = \int_0^1 k_\varepsilon(x, y') dy'$, the prototype being the Gaussian $k_\varepsilon(x, y) = e^{-(x-y)^2/\varepsilon}$.

Lemma 8.1 (Instant regularization). *For every $\varepsilon > 0$ and every $f \in \mathcal{K} \setminus \{0\}$, the smoothed profile $T_\varepsilon f$ is strictly positive on all of $[0, 1]$ (endpoints included) and non-increasing. Consequently d_H is well defined on every image $T_\varepsilon f$, even when f vanishes at an endpoint.*

Proof. Strict positivity: since $k_\varepsilon(x, y) > 0$ for all x, y and $f \geq 0$ with $f \not\equiv 0$, $(T_\varepsilon f)(x) = \int_0^1 \tilde{k}_\varepsilon(x, y) f(y) dy > 0$ for every $x \in [0, 1]$, the endpoints included. Monotonicity is the variation-diminishing property of TP2 kernels [15]: a TP2 kernel maps non-increasing functions to non-increasing functions, and row normalization (division by $Z_\varepsilon(x)$, which depends on x only) preserves the TP2 cross-ratio. The endpoint values, where an anchored profile of CDF type may vanish, are thereby lifted strictly above 0, so the Hilbert metric—defined only for $f \gg 0$ —applies to the entire image of the channel. $\square$

Proposition 8.2 (Explicit finite projective diameter of the Gaussian channel). *For $k_\varepsilon(x, y) = e^{-(x-y)^2/\varepsilon}$ the projective diameter of the image cone under T_ε is finite and computed exactly:*

$$\Delta(T_\varepsilon) = \sup_{x, y, u, v \in [0, 1]} \ln \frac{k_\varepsilon(x, y) k_\varepsilon(u, v)}{k_\varepsilon(x, v) k_\varepsilon(u, y)} = \frac{2}{\varepsilon}.$$

Proof. The Gaussian form gives

$$\ln \frac{k_\varepsilon(x, y) k_\varepsilon(u, v)}{k_\varepsilon(x, v) k_\varepsilon(u, y)} = -\frac{1}{\varepsilon} [(x-y)^2 + (u-v)^2 - (x-v)^2 - (u-y)^2] = \frac{2}{\varepsilon} (x-u)(y-v),$$

which on $[0, 1]^4$ attains its maximum at the corner $(x, u, y, v) = (1, 0, 1, 0)$, giving $\Delta(T_\varepsilon) = 2/\varepsilon < \infty$. $\square$

Definition 8.3 (Damped renormalization operator). With $\mathcal{K}$, $\mathcal{M}$, and the Gaussian smoothing channel T_ε as above, let

$$\mathcal{R}_{\eta, \varepsilon} := \mathcal{A} \circ ((1 - \eta) T_\varepsilon + \eta \Pi_{\text{ref}}),$$

where Π_{ref} is damping toward a reference profile with weight $\eta \in [0, 1]$ and $\mathcal{A}$ is an anchoring gauge that fixes the boundary normalization. Equip $\mathcal{K}$ with the Hilbert projective metric d_H .

Theorem 8.4 (Existence and uniqueness of the monotone fixed point). *For each fixed $\varepsilon > 0$, $\mathcal{R}_{\eta, \varepsilon}$ has a unique fixed point $Q^* \in \mathcal{M}$, and $d_H(\mathcal{R}^n f, Q^*) \leq \kappa(\varepsilon)^n d_H(f, Q^*)$ for all $f \in \mathcal{K}$, with $\kappa(\varepsilon) = \tanh(\Delta(T_\varepsilon)/4) < 1$.*

Proof. By Lemma 8.1 a single application of T_ε carries $\mathcal{M}$ into the strictly positive monotone profiles, where d_H is well defined; the dynamics may therefore be studied on this image. By Proposition 8.2 the channel has finite projective diameter $\Delta(T_\varepsilon) = 2/\varepsilon < \infty$, so by Birkhoff's contraction theorem [4, 7, 11] T_ε contracts d_H with coefficient $\tanh(\Delta/4) = \tanh(1/2\varepsilon) < 1$. The convex damping toward Π_{ref} does not increase d_H —it is a uniform additive (Doebelin) minorization, treated in Section 12—and the gauge $\mathcal{A}$ is d_H -invariant, selecting a representative within each projective class. Hence $\mathcal{R}_{\eta, \varepsilon}$ is a d_H -contraction with coefficient $\kappa(\varepsilon) = \tanh(\Delta(T_\varepsilon)/4)$. The cone of strictly positive monotone profiles is complete in d_H [7], so Banach's theorem [2] gives the unique fixed point Q^* and the geometric estimate; TP2 preservation (Lemma 8.1) keeps $Q^* \in \mathcal{M}$. $\square$

Lemma 8.5 (Nontriviality: the fixed point is a genuine transition). *Take the reference profile $\Pi_{\text{ref}}(x) = 1 - x$ and the gap functional $N(Q) := Q(0) - Q(1)$. Then the un-normalized damped profile satisfies $N((1-\eta)T_\varepsilon Q + \eta \Pi_{\text{ref}}) \geq \eta > 0$ for every $Q \in \mathcal{M}$. Consequently the anchoring gauge $\mathcal{A}$ never divides by a vanishing gap, and the fixed point Q^* is not the constant profile (which has gap 0): the bare Perron fixed point of the row-normalized smoothing alone—the constant—is excluded by the anchoring and the reference damping. Moreover, by the variation-diminishing property of the TP2 channel the curvature of Q^* has at most one sign change, so its density $-(Q^*)'$ is single-peaked and the inflection $\hat{a} \in (0, 1)$ of Definition 8.8 is interior and well defined.*

Proof. By Lemma 8.1 T_ε preserves monotonicity, so for nonincreasing Q the smoothed $T_\varepsilon Q$ is nonincreasing and $\tilde{N}(Q) := (T_\varepsilon Q)(0) - (T_\varepsilon Q)(1) \geq 0$. Hence

$$N((1-\eta)T_\varepsilon Q + \eta \Pi_{\text{ref}}) = (1-\eta)\tilde{N}(Q) + \eta N(\Pi_{\text{ref}}) = (1-\eta)\tilde{N}(Q) + \eta \geq \eta > 0,$$

using $N(\Pi_{\text{ref}}) = 1 - 0 = 1$. The gauge $\mathcal{A}$ normalizes by this gap, which is bounded below by η , so it is well defined and its image is non-constant; in particular $Q^* = \mathcal{R}_{\eta,\varepsilon}Q^*$ is non-constant. For the single interior maximum: the curvature measure of Π_{ref} vanishes (single-transition trivially), and a TP2 kernel does not increase the number of sign changes of a signed measure [15]; this bound is preserved along the d_H -contraction to Q^* , so the curvature of Q^* changes sign at most once and the density $-(Q^*)'$ is single-peaked. $\square$

Theorem 8.6 (Sharp exponential decay of the global contraction). *For the Gaussian kernel, $\Delta(T_\varepsilon) = 2/\varepsilon$, so*

$$\kappa(\varepsilon) = \tanh\left(\frac{1}{2\varepsilon}\right) = 1 - 2e^{-1/\varepsilon} + O(e^{-2/\varepsilon}), \quad 1 - \kappa(\varepsilon) \sim 2e^{-1/\varepsilon},$$

whereas the local diffusive spectral gap closes only polynomially, $1 - \lambda_1(\varepsilon) \sim C\varepsilon$.

Proof. The projective diameter of the Gaussian smoothing on monotone profiles is $\Delta(T_\varepsilon) = 2/\varepsilon$ (the log-ratio spread of the kernel over the unit interval), so by Theorem 8.4 $\kappa(\varepsilon) = \tanh(1/(2\varepsilon))$. Expanding $\tanh(z) = \frac{1-e^{-2z}}{1+e^{-2z}} = 1 - 2e^{-2z} + O(e^{-4z})$ at $z = 1/(2\varepsilon)$ gives $\kappa(\varepsilon) = 1 - 2e^{-1/\varepsilon} + O(e^{-2/\varepsilon})$. The periodic heat semigroup (standard normalization) has eigenvalues $e^{-4\pi^2 n^2 \varepsilon}$, so the local gap is $1 - e^{-4\pi^2 \varepsilon} \sim 4\pi^2 \varepsilon$, polynomial in ε . A change of scale convention rescales ε but leaves the qualitative contrast intact: global projective synchronization is exponentially harder than local smoothing. $\square$

Theorem 8.7 (Critical cooling law). *For the annealed iteration $Q_{n+1} = \mathcal{R}_{\varepsilon_n}(Q_n)$ with $\varepsilon_n = c/\ln n$, the cumulative contraction budget diverges iff $c \geq 1$:*

$$\sum_{n \geq 2} (1 - \kappa(\varepsilon_n)) = \infty \iff c \geq 1.$$

Proof. By Theorem 8.6, $1 - \kappa(\varepsilon_n) \sim 2e^{-1/\varepsilon_n} = 2e^{-\ln n/c} = 2n^{-1/c}$. The series $\sum_n n^{-1/c}$ diverges iff the exponent $1/c \leq 1$, i.e. $c \geq 1$ (p -series criterion). $\square$

Definition 8.8 (Inflection order parameter). The inflection of the fixed point is $\hat{a} := \arg \max_{x \in (0,1)} (-(Q^*)'(x))$, the location of maximal descent of the survival profile Q^* — equivalently the peak of its density $-(Q^*)'$, and the location of maximal slope of the complementary CDF $1 - Q^*$, the same point: the operator realization of the transition order parameter of Axiom A3.

Aggregation modes. Under loading, the micro-imperfection cascade activates in one of four modes, which fix the form class of Q^* : (i) parallel-instantaneous $\rightarrow$ logistic (central-limit aggregation); (ii) serial-cascaded $\rightarrow$ Weibull (weakest-link order statistics); (iii) multiaxial $\rightarrow$ hybrid (Hill/Gompertz); (iv) continuous-accumulating $\rightarrow$ Gaussian/logistic (diffusive averaging). A substrate is in general a convex combination on the simplex $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3, \pi_4) \in \Delta_4$; the mode identities are the basis families of the form-class vector space, and neither Theorem 8.4 nor the No-Go theorem below is affected by the mixture: only the inflection location and the mixture vector carry the substrate-specific asymmetry. The classical ductile/brittle dichotomy is the binary projection of this finer partition.

9 The No-Go Theorem

The central negative result: the inflection location, although well defined for each fixed point, is *not* a universal constant across the operator class. The mechanism is reparametrization invariance—a monotone coordinate change displaces any interior location—so the result is elementary as analysis; its force is *protective*: it forecloses the false reading of $\hat{a}$ as a natural constant and makes substrate-specific asymmetry the measured quantity rather than a defect.

Definition 9.1 (Reparametrization action and equivariant transition functional). The increasing homeomorphisms $h : [0, 1] \rightarrow [0, 1]$ act on profiles by $(h \cdot Q)(x) := Q(h(x))$, with $h^{-1} \cdot (h \cdot Q) = Q$. A class $\mathcal{G}$ of renormalization operators (TP2, strictly positive, anchored, damped) is *reparametrization-closed* if $R \in \mathcal{G}$ implies the conjugate $R^{(h)} := h^{-1} \circ R \circ h \in \mathcal{G}$ for every such h . A transition-location functional $\tau : \mathcal{M} \rightarrow (0, 1)$ is *equivariant* if $\tau(h \cdot Q) = h^{-1}(\tau(Q))$.

Theorem 9.2 (No-Go: no universal inflection constant). *Let $\mathcal{G}$ be reparametrization-closed and τ equivariant. Then there is no constant $c \in (0, 1)$ with $\tau(Q^*) = c$ for the fixed points Q^* of all $R \in \mathcal{G}$.*

Proof. Suppose such a c exists. Fix $R \in \mathcal{G}$ with fixed point Q^* , so $\tau(Q^*) = c$. Let $\tilde{c} \in (0, 1)$ be arbitrary and choose an increasing homeomorphism h with $h(c) = \tilde{c}$ (any smooth monotone h with $h(0) = 0, h(1) = 1, h(c) = \tilde{c}$ works). Form the conjugate $R^{(h)} = h^{-1} \circ R \circ h \in \mathcal{G}$ (reparametrization closure) and the profile $\tilde{Q}^* := h^{-1} \cdot Q^*$. Then

$$R^{(h)}(\tilde{Q}^*) = (h^{-1} \circ R \circ h)(h^{-1} \cdot Q^*) = h^{-1} \cdot R(h \cdot (h^{-1} \cdot Q^*)) = h^{-1} \cdot R(Q^*) = h^{-1} \cdot Q^* = \tilde{Q}^*,$$

so $\tilde{Q}^*$ is the fixed point of $R^{(h)}$. By equivariance, $\tau(\tilde{Q}^*) = \tau(h^{-1} \cdot Q^*) = h(\tau(Q^*)) = h(c) = \tilde{c}$. Since $\tilde{c}$ was arbitrary, the transition location ranges over all of $(0, 1)$ across $\mathcal{G}$, contradicting $\tau \equiv c$. $\square$

Corollary 9.3 (Symmetry recovers the constant). *A universal inflection location can hold only under a structural restriction that breaks reparametrization freedom. In particular, if the kernel of R is mirror-symmetric, the fixed-point profile satisfies the central symmetry $Q^*(x) = 1 - Q^*(1 - x)$, and its inflection (Definition 8.8) is at $1/2$.*

Proof. Let S be the central involution $(SQ)(x) := 1 - Q(1 - x)$. On the two-end-anchored survival profiles ($Q(0) = 1, Q(1) = 0$) it maps $\mathcal{M}$ to itself: SQ is nonincreasing with $(SQ)(0) = 1 - Q(1) = 1$ and $(SQ)(1) = 1 - Q(0) = 0$ (the bare domain reflection $Q \mapsto Q(1 - \cdot)$ reverses monotonicity; the range flip restores it). A mirror-symmetric kernel makes $\mathcal{R}$ commute with S , $\mathcal{R} \circ S = S \circ \mathcal{R}$; by uniqueness of the fixed point (Theorem 8.4), $Q^* = SQ^*$, i.e. $Q^*(x) = 1 - Q^*(1 - x)$. Differentiating gives $(Q^*)'(x) = (Q^*)'(1 - x)$, so the density is symmetric about $1/2$ and its unique maximum, the inflection of Q^* , lies at $x = 1/2$. The restriction to the centrally symmetric subclass is exactly the loss of reparametrization freedom required by Theorem 9.2. $\square$

Theorem 9.4 (Certified non-degeneracy under asymmetry). *For the damped Gaussian family parametrized by kernel asymmetry α , the inflection map $\alpha \mapsto \tau(Q_\alpha^*)$ is C^1 , and its first-order response at the symmetric point does not vanish. Writing*

$$\dot{Q}(1/2) := \partial_\alpha \Xi(\alpha)|_{\alpha=0} \in [1.8412, 1.8963] \neq 0$$

for the perturbation derivative of the symmetry-breaking functional Ξ — the first variation, at the midpoint, of the fixed-point condition under kernel asymmetry — this is a rigorous enclosure obtained by verified interval arithmetic at 256-bit precision (Krawczyk operator with Anselone bound and box enclosure) [13, 23]. Consequently $d\tau/d\alpha|_0 \neq 0$: the inflection shifts at first order off the symmetric point. The certified spectral bound $\rho(D\mathcal{R}) < 0.45$ guarantees geometric convergence at the certified parameters.

Proof. By the uniform contraction of Theorem 8.4 and the certified bound $\rho(D\mathcal{R}) < 0.45 < 1$, the linearization $I - D\mathcal{R}$ is invertible, so the implicit function theorem applied to $\mathcal{R}_\alpha(Q_\alpha^*) = Q_\alpha^*$ yields a C^1 branch $\alpha \mapsto Q_\alpha^*$, hence a C^1 map $\alpha \mapsto \tau(Q_\alpha^*)$. The first-order shift of the inflection is governed by the linearized inflection condition, whose coefficient is the perturbation derivative $\dot{Q}(1/2) = \partial_\alpha \Xi(\alpha)|_0$ (not the spatial slope of the profile); its enclosure $[1.8412, 1.8963]$ is produced by a reproducible ball-arithmetic certificate with auditable load-bearing steps. Since the enclosure excludes 0, $d\tau/d\alpha|_0 \neq 0$, so no neighbourhood of the symmetric kernel shares the inflection value $1/2$. (The inflection is non-degenerate: the symmetric fixed point has a strict density maximum, so the linearized inflection condition has nonzero leading coefficient $(Q^*)'''(1/2) \neq 0$ and $d\tau/d\alpha|_0 = -\dot{Q}(1/2)/(Q^*)'''(1/2)$ is well defined.) $\square$

Remark 9.5 (Scope). Theorem 9.2 is a non-existence statement and is complete as proved. Theorem 9.4 certifies one explicit asymmetric direction; the general statement that τ shifts for *every* admissible asymmetry, at all (ε, η) , remains an open conjecture, but the No-Go theorem does not require it: a single asymmetric kernel at which the inflection moves already refutes universality. The operative order parameter $\hat{a} = \arg \max_x (Q^*)'(x)$ (Definition 8.8) is itself a location functional: the inflection of a conjugate fixed point $Q^* \circ h$ is $\arg \max_x ((Q^*)'(h(x)) h'(x))$, which can be placed anywhere in $(0, 1)$ by choice of the increasing homeomorphism h , so no universal $\hat{a}$ -constant exists either; the abstract No-Go transfers to the concrete order parameter.

Conjecture 9.6 (The self-similar candidate Σ_c). A minimal self-similarity condition, whole-to-core as core-to-remainder, selects a candidate inflection: $\frac{1}{\Sigma} = \frac{\Sigma}{1-\Sigma} \iff \Sigma^2 + \Sigma - 1 = 0$, with positive root $\Sigma_c = \varphi - 1 = 1/\varphi \approx 0.618$. Three independent routes (the self-similarity condition, the damage fixed-point iteration $\Sigma_{t+1} = 1 - \Sigma_t^2$, and a dimensional optimization) converge on this root. This is a conjecture, not a theorem: by Theorem 9.2 no such value can be universal across $\mathcal{G}$, and the empirically observed inflections scatter in a structured, substrate-specific way around it. Proximity of an individual measured inflection to $1/\varphi$ is a datum, not support for universality.

10 The Second No-Go: No Universal Static Classifier under Interventional Monitoring

Theorem 9.2 eliminates universal *numerical* structure: no inflection constant survives reparametrization freedom. This section eliminates, at the deployment level of Section 4, universal *verdict* structure: no static binary classifier survives the closure of the measurement–action loop, once the published score enters the subsequent dynamics and the reception is adversive. The two results are projections of one limit picture: what is universal is the structural grammar and the monitoring architecture, never a constant and never a static verdict.

Definition 10.1 (Monitoring modes). Let (x_t) be the monitored trajectory and $\hat{a}_t$ the measured order parameter (Definition 4.1). Monitoring is *observational* if the dynamics do not depend on the measurement,

$$x_{t+1} = H_0(x_t),$$

and *interventional* if the published verdict enters the update,

$$x_{t+1} = H_I(x_t, \Phi(\hat{a}_t)),$$

for a classifier Φ with values in $\{0, 1\}$ ($1 = \text{UNSTABLE}$). In materials-testing terms the observational mode is non-destructive testing, the specimen is invariant under the measurement, while the interventional mode changes the boundary conditions of the loaded system through the act of reporting.

Definition 10.2 (Self-defeating reception). A system a_0 in a class $\mathcal{A}$ has *self-defeating reception* if its realized stability outcome negates every published verdict: writing $C(a_0) \in \{0, 1\}$ for the outcome under interventional monitoring,

$$C(a_0) = 1 - \Phi(a_0) \quad \text{for every static } \Phi.$$

The verdict UNSTABLE induces correction (the system stabilizes); the verdict STABLE induces complacency (the system destabilizes). This is the self-defeating prophecy [18], recorded here as a class property, not constructed from the axioms. We write

$$S_\Delta := \{a \in \mathcal{A} : a \text{ has self-defeating reception}\} \subseteq \mathcal{A}$$

for the subset of such systems.

Theorem 10.3 (Diagonal feedback barrier). *Let $\mathcal{A}$ be a system class with $S_\Delta \neq \emptyset$. Then no static classifier $\Phi : \mathcal{A} \rightarrow \{0, 1\}$ is correct on $\mathcal{A}$ under interventional monitoring.*

Proof. Correctness requires $\Phi(a) = C(a)$ for all $a \in \mathcal{A}$. For $a_0 \in S_\Delta$ this reads $\Phi(a_0) = 1 - \Phi(a_0)$, i.e. a fixed point of the negation on $\{0, 1\}$; the negation has none. $\square$

Remark 10.4 (Status of the theorem). Theorem 10.3 is the terminal step of the classical diagonal argument [16, 30]: fixed-point freeness of negation. It is not new mathematics, and the negation is *defined into* the class by Definition 10.2, not constructed. Its content is locative, parallel to Theorem 9.2: it fixes *where* the impossibility lives — in the closed, adverse measurement–action loop — and thereby what monitoring may still claim there. Whether a given real class (conversational trajectories under visible scoring among them) actually has $S_\Delta \neq \emptyset$ is an empirical premise: it can be supported by a controlled mode switch $H_0 \leftrightarrow H_I$, never settled formally.

Proposition 10.5 (A closed loop is necessary, not sufficient). *Let the reception of the published verdict be $\psi : \{0, 1\} \rightarrow \{0, 1\}$, so that the realized outcome at the loop is $C = \psi \circ \Phi$. Among the four self-maps of $\{0, 1\}$, the negation is the only fixed-point-free one. Consequently: if $\psi \neq \neg$ (in particular $\psi = \text{id}$, the self-fulfilling reception, and the constant receptions), then ψ has a fixed point b^* and the constant classifier $\Phi \equiv b^*$ is correct at the loop: a regime-bound classifier exists. Only the adverse loop $\psi = \neg$ raises the barrier.*

Proof. id fixes both points, the constants fix their value, and $\neg$ exchanges the two points. If $\psi(b^*) = b^*$, publishing b^* yields outcome $\psi(b^*) = b^*$, which is the published verdict. $\square$

Remark 10.6 (The $\Phi_{\mathcal{R}}$ diagnosis: two edges of one fixed-point picture). The existence question for a correct classifier is in both classical traditions a fixed-point question for the reaction map, answered at two different edges. On the continuous edge, a public prediction with *continuous* reaction admits a correct value by the intermediate-value/Brouwer argument, the classical possibility theorem of public prediction [12, 27]. On the binary edge, Proposition 10.5 locates the unique exception: only the adverse reception is fixed-point free. Continuity buys the fixed point; discreteness localizes its only failure.

The tempting strengthening, “a regime-bound classifier exists *iff* no reception coordinate is the negation”, holds only under *separability* (memoryless receptions acting coordinatewise, $\psi = (r_a)_a$). It fails in the coupled case: on $\{0, 1\}^2$ the rotation $\psi(x, y) = (y, 1-x)$ is fixed-point free although neither coordinate negates itself: of the nine fixed-point-free permutations of the four states, only three are separable. Adversiveness can be *distributed* across coupled axes (x follows y , y opposes x) without being attributable to any single axis. Since specimens carry loading-path memory — $\hat{a}_t$ is a functional of the whole history $x_{0:t}$ (Axioms A2, A4), and the deformation axes are coupled (Remark 4.2) — the non-separable case is the generic one for this framework. The characterization therefore remains a *diagnosis*, not a theorem: it unifies the two edges and names the exact residue (the sign of the coupled loop), which is an empirical question, not a structural one.

The design consequence is the operational counterpart of Theorem 9.2: monitoring must be specified as $\Phi_{\mathcal{R}}$ — bound to an explicit observation, feedback and intervention regime $\mathcal{R}$ — and never as a universal Φ_{univ} . This is the architecture already adopted in Definition 4.3: thresholds are calibration parameters of a regime, the state machine is the invariant. The continuous-optimization literature on prediction that shifts its own distribution [20] studies retraining equilibria for risk minimizers; the binary verdict cut and its barrier are a separate, elementary stratum beneath it.

Remark 10.7 (Two No-Gos, one picture). Theorem 9.2 forbids a universal *value* by reparametrization freedom of the substrate; Theorem 10.3 forbids a universal *static verdict* by fixed-point freeness of the adverse loop. Both eliminate universals in favour of structure: the functional form, the accumulation architecture, and the regime-bound measurement survive; constants and context-free classifications do not. The first No-Go makes the inflection a measured quantity; the second makes the verdict a regime property. Together they state the honest scope of the framework’s operational claim: structural grammar universal, numbers and verdicts local.

11 The Elimination Ledger

We collect, in reader-checkable form, what *cannot* be concluded at the present level of generality, and the minimal structure that would change each conclusion. These are not statements of empirical uncertainty: each is a consequence of the preceding theorems.

Proposition 11.1 (E1: no universal rate from local contractivity). *Per-step contractivity (existence of $\kappa_n < 1$) guarantees convergence once the budget diverges (Theorem 8.7), but does not determine a convergence-rate class (exponential / polynomial / ...) across systems: the rate is set by the cumulative budget $A_n := \sum_{k \leq n} (1 - \kappa_k)$, not by any single κ_n .*

Proof. By Theorem 8.4, $d_H(Q_n, Q^*) \leq (\prod_{k \leq n} \kappa_k) d_H(Q_0, Q^*) \leq e^{-A_n} d_H(Q_0, Q^*)$, using $\kappa_k \leq e^{-(1-\kappa_k)}$. The bound depends on A_n only; the example $\kappa_n = \tanh(1/(2\varepsilon_n))$ with $\varepsilon_n = c/\ln n$ (Theorem 8.7) gives $A_n \sim \sum n^{-1/c}$, whose growth class (and hence the rate) is tuned continuously by c while every local $\kappa_n < 1$. $\square$

Proposition 11.2 (E2: sharpness of budget bounds). *Even with an a-priori bound $d_n \leq e^{-A_n} d_0$, no universal improvement holds within the class: for any prescribed decay profile there is an admissible schedule (ε_n) realizing it.*

Proof. Given a target nonincreasing profile $\beta_n \downarrow 0$, choose κ_n with $\prod_{k \leq n} \kappa_k = \beta_n$ (i.e. $1 - \kappa_n = 1 - \beta_n / \beta_{n-1}$), realizable by $\varepsilon_n = 1 / (2 \operatorname{artanh} \kappa_n)$ since $\kappa \mapsto \tanh(1/(2\varepsilon))$ is a bijection onto $(0, 1)$. Each $\kappa_n < 1$ is admissible, so the decay can be made arbitrarily slow. $\square$

Proposition 11.3 (E3: schedule dependence). *When the per-step contraction degrades ($\kappa_n \rightarrow 1$, e.g. $\varepsilon_n \rightarrow 0$), convergence hinges on the explicit divergence $\sum_n (1 - \kappa_n) = \infty$ (Theorem 8.7); absent this schedule condition, persistent bad windows or non-convergence cannot be excluded.*

Proposition 11.4 (E4: no universal numerical threshold). *No concrete inflection/transition value (e.g. $1/2$, $1/\varphi$) is derivable at this level of generality; by Theorem 9.2 any such value requires structure that breaks reparametrization freedom.*

Proposition 11.5 (E5: no universal static classifier). *Correctness of a static classifier $\Phi : \mathcal{A} \rightarrow \{0, 1\}$ on a feedback-responsive class cannot be concluded without an empirically established reception sign for the deployed regime: if $S_\Delta \neq \emptyset$, no such Φ exists (Theorem 10.3); only $\Phi_{\mathcal{R}}$ -statements relative to a specified observation–feedback–intervention regime are admissible.*

Minimal structure that changes the conclusions.

(S1) Uniform minorization (Doebelin).

A genuine lower bound $Tf \geq c \langle f, \nu \rangle$ yields uniform mixing and a uniform rate (Section 12); converts E1/E3 into an explicit exponential rate.

(S2) Time-structure (good-step density / bounded gaps).

A lower bound on the frequency of contractive steps converts local contractivity into an explicit rate in terms of that density; addresses E1/E2.

(S3) Symmetry / identifiability.

Restriction to a mirror-symmetric or identifiable normal-form subclass pins a numerical inflection location (Corollary 9.3); addresses E4.

12 Spectral and Contraction Substructure

The rate of approach to the autopoietic fixed point is governed by the contraction budget, not by any local coefficient. We make the governing structure precise and exhibit the uniform-rate restoration (S1).

Definition 12.1 (Contraction budget). For a (possibly non-autonomous) iteration with per-step coefficients $\kappa_n \in (0, 1)$, the cumulative contraction budget is $A_n := \sum_{k \leq n} (1 - \kappa_k)$.

Theorem 12.2 (Budget governs convergence). $d_H(Q_n, Q^*) \leq e^{-A_n} d_H(Q_0, Q^*)$; in particular $A_n \rightarrow \infty$ implies $Q_n \rightarrow Q^*$, while a bounded budget leaves a non-vanishing contraction factor and so cannot force convergence. The asymptotic rate class is determined by the growth class of A_n and is invariant under any reparametrization that preserves A_n .

Proof. The product bound $\prod_{k \leq n} \kappa_k \leq e^{-\sum_{k \leq n} (1-\kappa_k)} = e^{-A_n}$ gives the estimate; if $A_n \rightarrow \infty$ then $d_H(Q_n, Q^*) \rightarrow 0$. Conversely, if A_n is bounded then $\prod \kappa_k \geq \prod e^{-2(1-\kappa_k)} = e^{-2A_\infty} > 0$ (using $\kappa \geq e^{-2(1-\kappa)}$ for κ near 1), so the contraction factor does not vanish and convergence to a unique limit cannot be forced. Reparametrization-invariance of the rate follows because A_n is built from the projective coefficients κ_k , which are gauge-invariant (Theorem 8.4). $\square$

Theorem 12.3 (Doebelin restart: uniform rate (S1)). *Replace T_ε by the regularized kernel $k_{\eta,\varepsilon} = (1-\eta)k_\varepsilon + \eta$ (a uniform minorization of weight η). Then the projective contraction satisfies $\kappa_\eta \leq \frac{1-\eta}{1+\eta} < 1$ uniformly in ε , and the annealed iteration converges provided $\sum_n \eta_n = \infty$, independently of the cooling schedule ε_n .*

Proof. A uniform additive component η is a Doebelin minorization $k_{\eta,\varepsilon} \geq \eta$; by the Doebelin coupling bound [10, 26] the Hilbert/total-variation contraction is at most $\frac{1-\eta}{1+\eta}$, with no dependence on ε . Hence $1 - \kappa_{\eta_n} \geq \frac{2\eta_n}{1+\eta_n} \geq \eta_n$, so the budget obeys $A_n \geq \sum_{k \leq n} \eta_k$; by Theorem 12.2, $\sum_n \eta_n = \infty$ forces convergence regardless of ε_n . $\square$

Remark 12.4 (Local versus global gap). Theorem 8.6 exhibits the central discrepancy: the *local* diffusive relaxation gap closes polynomially ($1 - \lambda_1 \sim C\varepsilon$), while the *global* projective synchronization gap closes exponentially ($1 - \kappa \sim 2e^{-1/\varepsilon}$). Local smoothing is easy; global ordering is exponentially hard. The certified spectral bound $\rho(D\mathcal{R}) < 0.45$ (Theorem 9.4) shows that at fixed, non-vanishing scale the realized linear gap is far wider than the Birkhoff-theoretic worst case: the degeneration is a singular-limit ($\varepsilon \rightarrow 0$) phenomenon, not a defect at operating scale. This is the spectral content behind the minimum-sampling requirement of the operational frame: a trajectory must accumulate enough budget for the inflection to be reconstructible.

Remark 12.5 (Closing). The framework is complete in the following sense. Sections 1–2 fix the class and reduce accumulation, transition and irreversibility to self-reference and dissipation. Section 3 transports the structure across scales. Section 4 instantiates it. Sections 5–7 prove the three structural theorems. Sections 8–9 construct the order-parameter fixed point and prove that its location is substrate-specific, not universal. Section 10 extends the elimination to the deployment level: under interventional monitoring with adversive reception no universal static classifier exists, and verdicts are regime-bound. Sections 11–12 delimit the rate theory. Every numerical inflection is, by Theorem 9.2, a measured quantity; what is universal is the structural grammar: the recurrence of one functional form, read off one cumulative observable, across substrates.

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